

VLCIM: A Vision-Language Cyclic Interaction Model for Industrial Defect Detection

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Abstract—Accurate defect detection is an important element in ensuring product quality and safe equipment operation. However, due to the lack of deep cross-modal interactions (CMIs) during vision feature extraction, existing methods often suffer from attention bias, which ultimately limits detection accuracy. To address this issue, this article proposes a vision-language cyclic interaction model (VLCIM), which progressively optimizes vision feature extraction by integrating domain prior knowledge and generic large model, effectively bridging the dual-domain barrier between “generic-specific” and “vision-language.” Specifically, progressive cyclic interaction learning is proposed for the first time, which integrates a recursive guidance module (RGM) and CMI strategy to realize bidirectional dynamic fusion and collaborative optimization of vision and language features. Furthermore, the proposed dual-view synergistic detection mechanism (DSDM) enhances discriminative decision responses, significantly improving the model’s boundary perception ability and decision-making accuracy in complex scenarios. VLCIM achieves high-precision defect detection by establishing a cyclic interaction mechanism between domain-specific language features and vision representations. The experimental results on three industrial datasets demonstrate that VLCIM achieves improvements of 5.9%, 5.6%, and 4.1% in mean intersection over union (mIoU) over the state-of-the-art (SOTA) methods, indicating its validity and generalization in different scenarios.

Index Terms—Defect detection, dual-view synergistic detection (DSD), industrial safety, large language model, progressive cyclic interactive learning (PCIL).

I. INTRODUCTION

DEFFECT detection is crucial in maintaining product quality and ensuring industrial system safety [1]. Traditional defect detection relies on experienced manual inspectors, which is not only time-consuming and labor-intensive but also difficult to meet the stringent requirements of modern industry

for high reliability and strong robustness [2]. In contrast, with the rapid development of vision technology, automatic inspection methods based on machine vision can effectively solve the problems of low efficiency, high labor intensity, and subjective bias in manual inspection, thus becoming a hot direction for research [3], [4], [5].

Early machine vision-based methods primarily relied on threshold segmentation, manually designed features, and shallow machine learning algorithms, which are widely applied in various industrial defect detection scenarios, such as steel strip, solar cell, and fabric. In [6] and [7], an adaptive threshold method and a guidance template-based method were proposed to realize strip steel surface defect detection. In [8], a Gaussian mixture model-based background subtraction method was proposed to extract the feature regions of weld defects (WDs). In [9], a cascade AdaBoost with a penalty term was proposed to realize pipeline defect detection. Although the above methods perform well in specific scenarios, they are reliant on pre-determined feature extraction rules, which undermines their viability in practical engineering applications.

In recent years, deep learning-based methods have attracted much attention in the field of industrial defect detection. Many excellent models based on convolutional neural networks (CNNs) have been widely used for the automatic extraction of complex defect features, which significantly improves the versatility and accuracy of detection [10]. Currently, deep learning-based defect detection methods are mainly categorized into three categories: image-level classification methods, region-level detection methods, and pixel-level segmentation methods [2]. Image-level classification methods [11], [12], [13], [14] mainly focus on identifying different classes of defects under a variety of working conditions. Region-level detection methods [15], [16], [17] roughly mark the location of defects through bounding boxes, but it is difficult to provide accurate boundary information. In contrast, pixel-level segmentation methods [18], [19], [20], [21] focus more on the precise classification of each pixel, and their goal is to capture the edges and details of the defects to achieve more comprehensive and accurate detection through fine-grained analysis. Although pixel-level segmentation methods have made significant progress in improving detection accuracy, they are generally limited by a single piece of vision information. When confronted with complex defects, these methods often struggle to effectively characterize features and accurately understand the context [22]. Therefore, the key to promoting further development of defect segmentation lies in constructing a brand new framework that can integrate richer and more comprehensive information resources, thus breaking

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TABLE I
COMPARISON OF THE RELATED STUDIES AND THE PROPOSED METHOD

Methods	Input	Interaction	Knowledge	Localization	Generalization	Complexity
Image-level classification	image	—	No	—	weak	low
Region-level detection	image	—	No	low	medium	medium
Pixel-level segmentation	image	—	No	medium	medium	high
Vision-Language model	image+text	unidirectional	No	medium	medium	high
Proposed method	image+text	cyclic	Yes	high	strong	high

through the bottleneck of existing methods and achieving superior detection results.

Recent advances in vision-language models, such as GPT [23], CLIP [24], GLIP [25], and VLPD [26], with their powerful image-text fusion capabilities, present promising opportunities to achieve this goal. Specifically, large-scale language models (LLMs) based on a large-scale corpus pretrained have accumulated rich generic prior knowledge, which can effectively enhance the feature representation of defects and improve the model's semantic cognition of defects [27]. However, the application of vision-language models in industrial defect segmentation is still in the exploratory stage due to the difficulty in acquiring industrial defect data and the large feature differences with generalized natural images. Notably, domain prior knowledge can provide inherent attributes of defects, such as shape and size, in the form of language, which usually do not vary with the scene. Therefore, this study aims to integrate the a priori knowledge from a large model with domain-specific knowledge and inject it into the vision encoder, ensuring that the model captures the most relevant feature representations of defects. However, such a process is by no means an easy task. The main challenges are attributed to two aspects: 1) at the feature level, how to realize effective interaction of multimodal information to ensure that language information can provide positive supervision of vision feature extraction and 2) at the decision level, how to guarantee the accuracy of segmentation decisions.

To address the first challenge, progressive circular interaction learning is first proposed to achieve iterative optimization of multimodal features. Specifically, domain-specific language information is constructed by combining a generalized large language model with expert knowledge. Inspired by the human progressive learning mechanism, a recursive guidance module (RGM) is designed to dynamically adjust the feature extraction behavior of the vision backbone by injecting language information layer by layer. Meanwhile, a cross-modal interaction (CMI) strategy is proposed to optimize the language embedding by using the deep-semantic information of images in reverse. The method significantly enhances the positive facilitation effect of language information to vision feature optimization through a progressive mutual supervision mechanism.

To address the second challenge, a dual-view synergistic detection mechanism (DSDM) is proposed. On the one hand, a decoder with a full multilayer perceptron (MLP) architecture is

proposed, which is designed not only to be highly coordinated with the Transformer-based encoder in terms of structure, but also to efficiently extend the receptive field through MLP to capture richer feature information. On the other hand, the complementary nature of the decision response is realized by constructing a dyadic hierarchy of forward and backward viewpoints (BVs). This mechanism significantly improves the model's localization accuracy and perception of complex defect boundaries, ensuring the accuracy and reliability of the overall decision-making.

Based on the above analysis, this article proposes a novel industrial defect detection method named vision-language cyclic interaction model (VLCIM). The method guides the vision encoder layer by layer through language information, focusing on key vision features related to language descriptions. Meanwhile, the model can integrate the deep semantic information of defects to further enhance the expressive effectiveness of language information. In addition, the DSDM effectively improves the accuracy and reliability of the detection results. In addition, to show the advantages of the proposed method more clearly, Table I provides a comparative overview of the mainstream defect detection methods. Noteworthy, unlike traditional vision-language models, VLCIM realizes a good balance between generalization, boundary localization, and computational efficiency through the proposed cyclic interaction strategy and a DSDM. Overall, the contributions of this article can be summarized as follows.

- 1) This study explores the effective application of vision-language model in industrial defect detection tasks and breaks through the dual-domain barrier between "generic-specific" and "vision-language" by incorporating expert experience and the concept of progressive cyclic, providing a new solution to industrial defect detection.
- 2) A VLCIM is proposed, integrating progressive cyclic interaction learning and a DSDM. The former leverages domain-specific language information to guide the vision encoder in progressively focusing on defect-related vision features, while employing vision semantic information to supervise language embeddings in reverse. The latter ensures precise defect segmentation by utilizing complementary perspectives to enhance decision-making accuracy.
- 3) The experimental results on three industrial datasets demonstrate the effectiveness of the proposed method.

On the NEU-Seg, CVC-Seg, and magnetic field leakage (MFL)-Seg datasets, the intersection over union (IoU) of this method improved by 5.9%, 5.6%, and 4.1%, respectively, compared to advanced methods, thus establishing the state-of-the-art (SOTA) results.

The remainder of this article is organized as follows. Sections II and III describe the related works and the proposed methodology, respectively. Section IV presents the experimental results and their analysis. Finally, Section V summarizes this article.

II. RELATED WORKS

This section provides a brief review of prior work on deep learning-based defect detection methods, encompassing image-level defect classification, region-level defect detection, and pixel-level defect segmentation. Additionally, related studies on large vision-language models are discussed.

A. Deep Learning-Based Defect Detection Methods

1) *Image-Level Defect Classification*: The image-level defect classification aims to identify different patterns of defect. In [12], an integrated CNN model via long and short-term memory is designed for highly accurate recognition of fabric defects. In [13], a consistent multiscale feature mapping method was proposed to realize defect recognition of low-quality weld images. In [14], an inception-CNN-based model was designed to implement servo motor surface defect recognition, which effectively captures multiscale local information by introducing inception modules between convolutional layers. Although the above methods have the advantages of flexibility and computational efficiency, these methods are difficult to accurately localize the defect location, and the recognition performance tends to degrade when multiple defects appear at the same time.

2) *Region-Level Defect Detection*: Region-level detection methods utilize end-to-end detectors such as Faster R-CNN [28], and You Only Look Once (YOLO) [29] and their variants to realize defect detection. In [15], a cascaded attention fusion network was proposed to mine explicit and implicit features of defects through complementary attention mechanisms to achieve complex tube body defect detection. In [16], an improved YOLO framework for surface defect detection on hot rolled strip steel was proposed by combining a Swin Transformer and adaptive spatial feature fusion. In [17], a fast and efficient detection network was designed to achieve surface defect detection, which addresses the challenges of large texture variations and small defect sizes by incorporating the feature retention block and the skip-dense connection. However, the above methods can usually only locate the general area of the defect, and it is difficult to accurately delineate its boundaries, especially when facing irregularly shaped defects.

3) *Pixel-Level Defect Segmentation*: Pixel-level defect segmentation aims to classify each pixel to accurately identify and localize defective regions. In [18], an improved U-Net was designed to implement radiator surface defect detection, aggregating a lightweight convolutional module and a self-attention mechanism. In [19], Deeplab with multiscale attention was

proposed to realize pavement crack segmentation where a multiscale attention module was used to generate an attention mask for the dynamic fusion of different feature mappings. In [20], a dual-branch semantic segmentation network was designed to realize surface defect detection, which combines semantic information and boundary information to enrich the feature representation of defects. Despite the excellent performance achieved by the above methods, their reliance on single modal information limits the generalization ability of the models. To this end, this article introduces linguistic modality to enhance the ability to model the semantics of defects and further improve segmentation accuracy and generalization.

B. Large Vision-Language Model

The emergence of LLMs, such as GPT [23], marks the arrival of a new era in artificial intelligence. These models are trained on large-scale data, incorporate rich prior knowledge, and provide powerful support for a wide range of tasks. Recently, researchers have begun to explore the vision-language model, hoping to build a unified multimodal data processing framework to make up for the limitations of LLM in processing nonlanguage data [24], [25], [26], [27]. In [24], a large-scale pretrained model called CLIP was proposed, which leverages a contrastive learning strategy to train on massive image-text pair datasets and demonstrates outstanding transferability across 30 classification datasets. Building on the inspiration from this work, many vision-language models have been proposed for applications in areas such as object detection [25] and semantic segmentation [26], [27]. Currently, there are fewer studies on industrial defect detection based on vision-language model. Compared to natural images, industrial data have more complex features and domain-specific knowledge, and direct pretrained with generalized large models may lead to poor performance. Therefore, this study combines expert experience and generic large models to obtain domain-specific language representations. Meanwhile, unlike existing language-embedded methods, progressive circular interaction learning is proposed for the first time to realize bidirectional supervision and joint optimization of vision-language features.

III. METHODOLOGY

A. Methodology Overview

As shown in Fig. 1, considering the advantages of language information in conveying expertise, a novel vision-language model called VLCIM is proposed to realize accurate surface defect detection. The model incorporates language priors to guide the learning process and employs a progressive cyclic interaction design to achieve bidirectional iterative optimization of multimodal information. In the *multimodal feature extraction* (MFE) phase, text and images are, respectively, fed into the CLIP language encoder and image encoder, with expert correction (EC) applied to extract the initial multimodal features. In the *progressive cyclic interaction learning* phase, the designed RGM enables the gradual modulation of multi-stage vision features by language features. Additionally, CMI is designed to realize reverse supervision and optimization of language features. In the dual-view synergistic detection

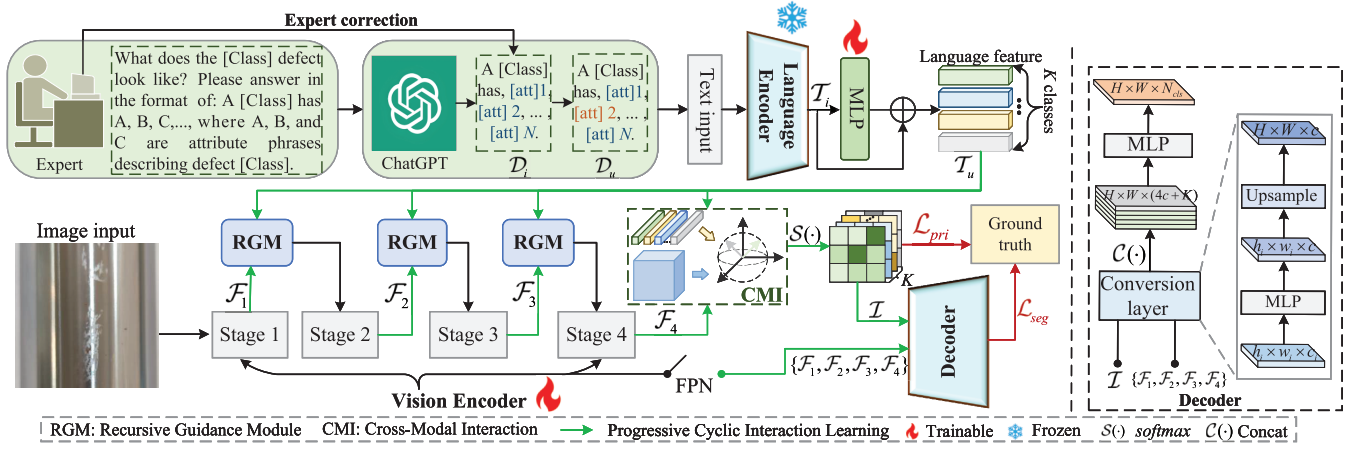


Fig. 1. Proposed VLCIM framework. It consists of five components: 1) language encoder, which extracts linguistic features from textual descriptions; 2) RGM, which generates dynamic weights and biases for the vision encoder based on linguistically-specific information; 3) vision encoder, which extracts vision features from the original image, and whose attention can be modified by linguistically adaptive weights; 4) CMI, which optimizes the language features based on deeper vision information; and 5) decoder, which generates predictions.

(DSD) phase, the optimized multimodal features are fed into the decoder to realize surface defect detection. The details are as follows.

B. Multimodal Feature Extraction

MFE is applied to extract defect-related language features and vision features. Specifically, the Transformer structure in CLIP is applied to the multimodal encoders. In particular, the language information is corrected via expert experience to reasonably embed the domain knowledge. Define the multimodal dataset as $\Omega = \{\mathcal{X}_j, \mathcal{G}_j, \mathcal{D}_j\}_{j=1}^N$, where $\mathcal{X}$ and $\mathcal{G}$ are the input and corresponding labels, respectively. $\mathcal{D}$ is the corresponding text description. N denotes the number of samples. Details are as follows.

1) *Language Feature Extraction*: Existing studies suggest that detailed and consistent text descriptions of vision images can lead to more accurate language features. Therefore, different from a single category prompt from the template “This is a photo of a [Class]” with K class names, this article provides a more complete description of language attributes through Chat-GPT [30] and expert experience to bridge the domain gap between language and vision. As shown in Fig. 1, an expert issues a command to Chat-GPT “what does the [viscose] defect look like?” and then obtains an initial language attribute description “ $\mathcal{D}_i = \{\text{A viscose has irregular texture, size deviation, and uneven thickness}\}$.” It is worth noting that, considering the differences between the defects in specific conditions and generic scenarios, the expert further corrects the $\mathcal{D}_i$ to filter out the wrong attributes and update domain-specific attributes. Therefore, the updated text input “ $\mathcal{D}_u = \{\text{A viscose has irregular texture, cracks and creases, and inconsistent color}\}$ ” is obtained, which fully guarantees the accuracy of the text information. Next, $\mathcal{D}_u$ is sent to a frozen language encoder to obtain the initial language features $\mathcal{T}_i \in \mathbb{R}^{K \times d}$, where d denotes the feature dimension. Inspired by low-rank adaptation [31], an adaptive layer (AL) $\mathcal{M}(\cdot)$ is built via an MLP for fine-tuning the initial features $\mathcal{T}_i$ to be consistent with the scene-specific image input in $\mathcal{T}_u = \mathcal{T}_i + \eta \mathcal{M}(\mathcal{T}_i, \vartheta)$,

where η denotes the tradeoff coefficient and is set to 0.01 to maximize the retention of the original valid language information. ϑ and $\mathcal{T}_u$ denote the parameter of MLP and the updated language features, respectively.

2) *Vision Feature Extraction*: As shown in Fig. 1, the vision encoder $\Psi(\cdot)$ includes four stages, where each stage consists of multiple Transformer subblocks. When image $\mathcal{X}_i$ is sent into the vision encoder, the initial vision features $\{\mathcal{F}_i\}_{i=1}^4 = \Psi(\mathcal{X}_j, \theta)$ can be obtained, where θ denotes the parameters of the vision encoder. Notably, existing studies [32], [33], [34] have shown that the features extracted from the last layer of the CLIP vision encoder are rich in semantic information (i.e., language-compatible features), aligning well with the language features. Therefore, the optimization of language features is performed based on the deep semantic features $\mathcal{F}_i$.

C. Progressive Cyclic Interactive Learning

To fully realize the deep modulation of vision features by language information, unlike previous language-guided methods, a progressive cyclic interaction learning method is proposed for the first time, enabling mutual supervision and joint optimization of language and vision features. Specifically, the language priors encoded by the language encoder are considered effective in guiding the model to attend to critical vision features, thereby enhancing the accuracy of vision understanding. Conversely, deep semantic features refine language features in a feedback manner, making them more aligned with the requirements of vision tasks. The proposed method comprises two parts including language-guided vision refinement and vision-driven language optimization.

1) *Language-Guided Vision Refinement*: Conditional batch normalization (CBN) [35] utilizes language information to generate two feature affine transformation parameters within a single normalization layer for modulating the vision feature. Therefore, the optimized feature $\mathcal{F}_i$ obtained through the two affine parameters is calculated in the following equation:

$$\mathfrak{J}(\mathcal{F}_i, \xi, \varpi) = \xi \mathcal{F}_i + \varpi \quad (1)$$

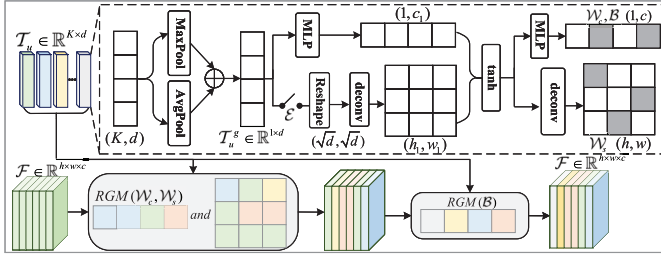


Fig. 2. Overview of the proposed RGM, where the dotted line shows the detailed structure of the RGM.

where $\mathfrak{I}(\cdot)$ denotes the operation of CBN. ξ and ϖ denote the learnable affine transformation parameters. However, CBN lacks sufficient flexibility because it is confined between the convolution and the activation function. In addition, CBN focuses only on the aggregation of channel information and ignores spatial information, which is crucial for defect localization. Therefore, to more fully realize the guidance of language information, an RGM is designed, which generates channel weights and spatial weights via language features and fuses them with vision features so as to progressively direct the vision encoder's attention to the regions associated with language descriptions.

As shown in Fig. 2, for the vision feature $\mathcal{F} \in \mathbb{R}^{c \times h \times w}$, the RGM dynamically generates a set of adaptive weights including channel weights $\mathcal{W}_c \in \mathbb{R}^{1 \times c}$ and spatial weights $\mathcal{W}_s \in \mathbb{R}^{h \times w}$, and biases $\mathcal{B} \in \mathbb{R}^{1 \times c}$. For channel weights and biases, first, maximum pooling $\mathcal{P}_{\max}(\cdot)$ and average pooling $\mathcal{P}_{\text{avg}}(\cdot)$ are applied to capture global information $\mathcal{T}_u^g \in \mathbb{R}^{1 \times d}$ about language features. Then, the MLPs are applied to map $\mathcal{T}_u^g \in \mathbb{R}^{1 \times d}$ to the same dimensions as the channels of vision features in the following equation:

$$\mathcal{Z}_c = \text{MLP}_c^1(\mathcal{T}_u^g, \ell_1), \quad \mathcal{Z}_b = \text{MLP}_b^1(\mathcal{T}_u^g, \eta_1) \quad (2)$$

where ℓ_1 and η_1 denote the parameters of different MLPs, respectively. $\mathcal{Z}_c$ and $\mathcal{Z}_b$ denote the initial channel weights and channel biases, respectively. Next, the tanh activation function and another MLP layer are employed to obtain the final channel weights $\mathcal{W}_c \in [-1, 1]$ and channel biases $\mathcal{B} \in [-1, 1]$ in the following equation:

$$\mathcal{W}_c = \text{MLP}_c^2(\mathcal{Z}_c, \ell_2), \quad \mathcal{B} = \text{MLP}_b^2(\mathcal{Z}_b, \eta_2) \quad (3)$$

where ℓ_2 and η_2 denote the parameters of another MLP.

For spatial weights, similar to the above process, a global feature representation $\mathcal{T}_u^g \in \mathbb{R}^{1 \times d}$ is obtained. Previous studies focus on utilizing broadcast mechanisms or convolution to achieve dimensionality expansion, which not only increase some parameters but also potentially compromises the spatial location information. Unlike previous studies, the positional encoding $\mathcal{E}$ is applied to $\mathcal{T}_u^g$ and further the language feature $\mathcal{T}_u^g$ is reshaped into a 2-D matrix, which maximizes the preservation of its spatial information. Next, two deformable convolutional layers $\text{Dec}(\cdot)$ and tanh activation function are applied to generate spatial weights in the following equation:

$$\mathcal{W}_s = \text{Dec}_2(\tanh(\text{Dec}_1(\mathcal{R}(\mathcal{T}_u^g, \mathcal{P}), \hat{h}_1), \hat{h}_2)) \quad (4)$$

where $\hat{h}_1$ and $\hat{h}_2$ denote the parameters of the deformable convolution. Finally, as expressed in the following equation,

the weights and biases are applied to replace the affine transformation parameters ξ and ϖ in (1), enabling fine-grained control over the activation of each stage in the vision encoder:

$$\mathfrak{I}(\mathcal{F}_i | \mathcal{W}_c, \mathcal{W}_s, \mathcal{B}) = \mathcal{W}_s(\mathcal{W}_c \mathcal{F}_i) + \mathcal{B}. \quad (5)$$

The above weights and biases are limited to the range of -1 to 1 , and their values reflect the importance of each channel and spatial region in the vision features. Leveraging this mechanism, RGM can selectively reinforce the defect features based on language information, gradually guiding the vision encoder to focus on the defect regions.

2) *Vision-Driven Language Optimization*: In this section, a CMI mechanism is designed that exploits the relationship between language and pixels to optimize language embedding. Specifically, deep vision features $\mathcal{F}_4$ and language features $\mathcal{T}_u^g$ are applied to compute the relational interaction map $\bar{\mathcal{I}} \in \mathbb{R}^{(h \times w) \times K}$ in the following equation:

$$\bar{\mathcal{I}} = \text{Cos}(\lambda(\mathcal{F}_4), (\lambda(\mathcal{T}_u^g))), \quad \text{for } i \text{ in } [1, K] \quad (6)$$

where $\text{Cos}(\cdot)$ and $\lambda(\cdot)$ denote the cosine similarity and the normalization along the channel dimension, respectively. Then, the *softmax* activation function $\mathcal{S}(\cdot)$ with a temperature τ is adopted to normalize interaction map $\bar{\mathcal{I}} \in [-1, 1]$ to $\mathcal{I} \in [0, 1]$ in the following equation:

$$\mathcal{I} = \mathcal{S}(\bar{\mathcal{I}}) = \left\{ \frac{\exp(\bar{\mathcal{I}}_i^a / \tau)}{\sum_{d \in K} \exp(\bar{\mathcal{I}}_i^d / \tau)} \mid a \in K, i = 1, 2, \dots, hw \right\}. \quad (7)$$

The updated interaction map $\mathcal{I}$ illustrates the alignment between language and image, serving as the key basis for the backward optimization of the language model. Specifically, $\mathcal{I}$ can be regarded as K low-resolution language prior segmentation result. By calculating the loss $\mathcal{L}_{\text{pri}}$ between $\mathcal{I}$ and the ground truth $\mathcal{G}$, the domain discrepancy between language features and vision features can be further reduced in the following equation:

$$\mathcal{L}_{\text{pri}} = -\frac{1}{N} \sum_{n=1}^N (\mathfrak{N}(\mathcal{I}_n) \log \mathcal{G}_n + (1 - \mathfrak{N}(\mathcal{I}_n)) \log(1 - \mathcal{G}_n)) \quad (8)$$

where $\mathfrak{N}(\cdot)$ denotes the operation of upsampling and $N = H \times W$. H and W denote the height and width of the input, respectively. In addition, inspired by PSPNet [36], the interaction map $\mathcal{I}$ is concatenated to corresponding vision feature $\mathcal{F}_4$ to explicitly merge language priors and form the jointed features $\mathcal{F}_4 \leftarrow [\mathcal{F}_4, \mathcal{I}]$. Finally, the jointed features and the interaction map are fed together into the decoder to realize defect detection. Notably, the feature pyramid network (FPN) is applied to guarantee the accuracy of multiscale detection.

D. Dual-View Synergistic Detection

Existing studies rely heavily on single-perspective decision-making, which often fails to realize satisfactory accuracy for boundary localization of complex defects. To this end, this study proposes a DSDM, which utilizes the complementary nature of the two perspectives to learn different information from target and nontarget. In addition, the decoder contains

only the MLP layer, which not only maintains a high degree of consistency with the Transformer-driven encoder in terms of architectural design, but also gives full play to the advantages of MLP in global feature modeling. By utilizing the powerful global modeling capability of MLP, more global conlanguage information can be effectively captured, which provides a strong guarantee for improving the accuracy of defect detection.

Algorithm 1 VLCIM Training Process

Input: Multimodal dataset: $\Omega = \{\mathcal{X}_j, \mathcal{G}_j, \mathcal{D}_j\}_{j=1}^N$, the number of iterations T .

Output: The fully trained VLCIM parameters.

- 1: Initialize: Parameters of language encoder, vision encoder, and decoder: $\mathbb{F}_l(\theta)$, $\mathbb{F}_v(\theta)$, and $\mathbb{F}_d(\zeta)$.
 - 2: **for** iteration = 1 to T **do**
 - 3: Input text description $\mathcal{D}_j$ to $\mathbb{F}_l(\theta)$, extract the language feature $\mathcal{T}$;
 - 4: Input language feature $\mathcal{T}$ and image $\mathcal{X}_j$ to $\mathbb{F}_v(\theta)$, obtain the modulated vision features $\mathcal{F}$;
 - 5: Calculate the priori loss $\mathcal{L}_{pri}$ according to Eq. (8);
 - 6: Input $\mathcal{I}$ and $\mathcal{F}$ to decoder, obtain the dual-view predictions: $\{\mathcal{H}_f, \mathcal{H}_b\} = \mathbb{F}_d(\mathcal{I}, \mathcal{F}|\zeta)$
 - 7: Apply $\{\mathcal{H}_f, \mathcal{H}_b\}$ and $\mathcal{G}$ to calculate the dual-view loss $\mathcal{L}_f$ and $\mathcal{L}_b$;
 - 8: # Optimize the parameters of VLCIM
 - 9: Minimizing Eq. (14) with AdamW;
 - 10: **end for** **return** the optimal model VLCIM.
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As shown in Fig. 1, multiscale features $\mathcal{F}$ and the interaction map $\mathcal{I}$ are fed into the transition layer consisting of MLPs to realize channel dimension alignment in the following equation:

$$\begin{cases} \tilde{\mathcal{F}}_i = \text{MLP}_d^i(\mathcal{F}_i, v_i), & \text{for } i \text{ in } [1, 3] \\ \tilde{\mathcal{F}}_4 = \text{MLP}_d^4(\mathcal{C}(\mathcal{F}_4, \mathcal{I}), v_4) \end{cases} \quad (9)$$

where v denotes the parameters of different MLPs. Then, upsampling is applied to transform the aligned features $\tilde{\mathcal{F}}_i \in \mathbb{R}^{h_i \times w_i \times c}$ into $\tilde{\mathcal{F}}_i \in \mathbb{R}^{H \times W \times c}$. Next, the features $\tilde{\mathcal{F}}_i$ are concatenated and another MLP is applied to obtain the predicted features $\tilde{\mathcal{Q}} \in \mathbb{R}^{H \times W \times 4c}$ in the following equation:

$$\tilde{\mathcal{Q}} = \text{MLP}(\mathcal{C}(\tilde{\mathcal{F}}_1, \tilde{\mathcal{F}}_2, \tilde{\mathcal{F}}_3, \tilde{\mathcal{F}}_4), v) \quad (10)$$

where $\mathcal{C}(\cdot)$ denotes the operations of connection. Subsequently, the details of forward viewpoint (FV) prediction and BV prediction are elaborated as follows. For *FV prediction*, the model should be guided to focus on learning the response of the foreground categories, where the pixels denote defects. Specifically, the forward response $\mathcal{H}_f \in \mathbb{R}^{H \times W \times N_{cls}}$ is obtained through another MLP and the prediction feature $\mathcal{Q}$, where N_{cls} denotes the number of classes of defects. Next, the loss function $\mathcal{L}_f$ is calculated between ground truth $\mathcal{G}$ and forward decision response $\mathcal{H}_f$ in the following equation:

$$\mathcal{L}_f = -\frac{1}{N} \sum_{n=1}^N (\mathcal{H}_f^n \log \mathcal{G}_n + (1 - \mathcal{H}_f^n \log (1 - \mathcal{G}_n))). \quad (11)$$

For *BV detection*, the network should focus on learning the responses unrelated to the foreground categories (i.e., pixels

that are not defects), thereby indirectly enhancing the model's ability to distinguish defect regions, particularly the boundaries of the defects. Unlike the FV prediction, an inverse activation function $\mathcal{R}(\cdot)$ is applied to the output of the MLP layer to obtain the backward response $\mathcal{H}_b \in \mathbb{R}^{H \times W \times N_{cls}}$ in the following equation:

$$\mathcal{H}_b = \frac{1}{\text{ReLU}(\text{MLP}(\mathcal{Q}, v_b)) + 0.125} - 4. \quad (12)$$

Then, the loss function $\mathcal{L}_b$ is calculated between ground truth $\mathcal{G}$ and backward decision response $\mathcal{H}_b$ in the following equation:

$$\mathcal{L}_b = -\frac{1}{N} \sum_{n=1}^N (\mathcal{H}_b^n \log \mathcal{G}_n + (1 - \mathcal{H}_b^n \log (1 - \mathcal{G}_n))). \quad (13)$$

Finally, the overall optimization loss $\mathcal{L}$ is defined in the following equation:

$$\mathcal{L} = \kappa \mathcal{L}_{pri} + (1 - \kappa)(\mathcal{L}_f + \mathcal{L}_b) \quad (14)$$

where κ denotes the tradeoff coefficient and is set to 0.2 and 0.3 in different datasets, respectively. Noteworthy, in the inference process, the output of the FV is applied as the final result. Algorithm 1 summarizes the training process. The complementary interaction between the two viewpoints enables the model to more comprehensively capture the details of the defect region, thereby significantly improving the accuracy of defect detection.

IV. EXPERIMENTAL ANALYSIS AND DISCUSSION

To evaluate the effectiveness of the proposed method, three industrial defect datasets are applied including the hot-rolled steel strip defect dataset (NEU-Seg), the control valve core surface defect dataset (CVC-Seg), and the pipeline magnetic flux leakage defect dataset (MFL-Seg). A comparative study is conducted using several classical and SOTA methods, including GSLU-Net [18], PSPNet [36], DNLNet [37], CTNet [38], PIDNet [39], DMANet [19], ISANet [40], CRNet [21], and DDSNet [20]. To ensure fairness in the experiments, the parameters for the comparison methods are carefully adjusted to align as closely as possible with their original optimal settings. Notably, each experiment is repeated five times, and the average results are reported.

A. Dataset Description

1) *Neu-Seg Dataset*: The NEU-Seg dataset is a standardized high-quality database collected by Song and Yan [41] to solve the problem of automatic identification of defects in the hot-rolled strip. Due to the complexity of the hot-rolled sheet condition, there are large differences in the appearance of defects within classes, and defects between classes contain similar parts with low contrast to the background. The segmented dataset includes three typical types of defects: inclusion (IN), patch (PA), and scratch (SC), which is acquired in a light environment and contains a total of 4170 images with a resolution of 200×200 . The training and test sets contain 3630 and 840 images, respectively, and are labeled pixel by pixel using the open labeling tool Labelme.



Fig. 3. CVC-Seg dataset collection platform. (a) Control valve and valve core production platforms. (b) Display of valve core.

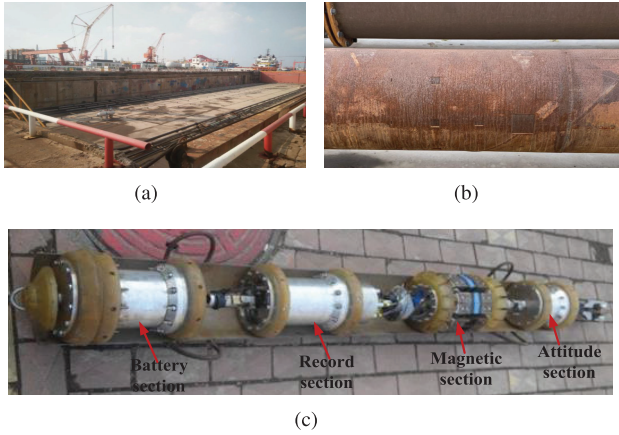


Fig. 4. Experimental site and MFL inspection robot. (a) Pipeline platform. (b) Actual defects. (c) MFL inspection robot.

2) *CVC-Seg Dataset*: As shown in Fig. 3, the CVC-Seg dataset is collected from the Northern China control valve production platform. There are four common types of defects found in the manufacturing process: viscose (VI), SC, indentation (IND), and pullmark (PM). Before the valve core is installed, it must be inspected and measures taken via the type of defect: 1) valve cores without any damage are stored in the warehouse; 2) those with adhesive residue are directed to the cleaning process for VI removal; 3) valve cores exhibiting surface SCs are moved to the scrap area, where their packaging is reinforced; and 4) cores with pull marks are also sent to the scrap area, and adjustments are made to the valve core production process. Affected by the line-axis mirroring effect, the CVC-Seg dataset is collected under a low-light environment and contains 900 images, each with a resolution of 1400×1400 , and the contrast between the defects and the background is low. The dataset is labeled at the pixel level and divided into a training set and a test set in the ratio of 7:3, and the samples in the two subsets do not overlap at all.

3) *MFL-Seg Dataset*: The MFL-Seg dataset can reflect the corrosion condition of the pipeline, and the experimental data are collected from an actual pipeline platform, as shown in Fig. 4. The MFL signal is acquired by an inspection robot, which contains magnetic section, battery section, attitude section, and record section. During the working process, the robot is propelled by oil and gas drive inside the pipeline, the operating pressure is 3 MPa, the speed range is 0–5 m/s, and the sampling interval is 0.002 m. The raw data are 2-D MFL signals, which are converted into images to present the defect features more clearly by referring to the work. The

MFL-Seg dataset contains a total of 841 pixel-level labeled samples with a resolution of 308×432 , and the categories include spiral weld (SW), girth weld (GW), and WD. The dataset is randomly divided into training and test sets in the ratio of 7:3 with no sample overlap between the two.

B. Experimental Setting and Evaluation Metrics

1) *Network and Hardware Setting*: The MFE structure of the proposed method is all based on the Transformer and its optimizer is Adam weight decay (AdamW) with a weight decay factor of 0.0001. The initial learning rate is 0.0001, and the learning rate is optimized in poly. The batch size is set to 4. The maximum number of epochs is 200. All the codes are implemented in the mmsegmentation framework, and different data enhancement methods including random clipping and random rotation are applied to reduce overfitting. The hardware setup consists of a 13th generation Intel¹ Core² i9-13900K CPU, 128 GB of RAM, and an NVIDIA GeForce RTX 4090 GPU, and the operating system is Ubuntu 22.04.

2) *Evaluation Metrics*: In the experiment, Precision rate, i.e., $\text{Precision} = (\text{TP}/(\text{TP} + \text{FP})) \times 100\%$, Recall rate, i.e., $\text{Recall} = (\text{TP}/(\text{TP} + \text{FN})) \times 100\%$, and *F1*-score, i.e., $F1 = (2 \times \text{Pre} \times \text{Rec})/(\text{Pre} + \text{Rec}) \times 100\%$ are applied to evaluate the detection performance of the model, where TN, FP, FN, and TP denote the true negative, false positive, false negative, and true positive, respectively. Specifically, the Precision rate and Recall rate can indicate the misdiagnosis and missed diagnosis of the model. *F1*-score can verify the overall performance of the model more comprehensively. In addition, $\text{IoU} = (O_p \cap O_g)/(O_p \cup O_g) \times 100\%$ is applied to measure the localization performance of the model, where O_g and O_p denote the label set and prediction set, respectively. For a specific category, $O_p \cap O_g$ denotes the common pixels shared by both the predicted and labeled sets in that category, while $O_p \cup O_g$ denotes all pixels that belong to either the predicted or the labeled sets in that category. In addition, to evaluate the performance of the model as a whole, mean Precision (mP) and mean IoU (mIoU) are applied.

C. Comparison With SOTAs

In this experiment, the detection accuracy of the proposed method is compared with that of other SOTA methods. In addition, a comparison of the detection results of different methods is given. The experimental results and analysis are as follows.

1) *Neu-Seg Analysis*: The detection comparison results of the proposed method with other methods on the strip surface defect images are shown in Fig. 5. From Fig. 5, it can be observed that the proposed VICIM outperforms the other methods in challenging defect detection scenarios such as low contrast (rows 1 and 2) and intraclass variation (row 3), and its prediction results are highly consistent with the ground truth. In addition, the quantitative comparison results, as shown in Table II, show that the proposed method outperforms other

¹Registered trademark.

²Trademarked.

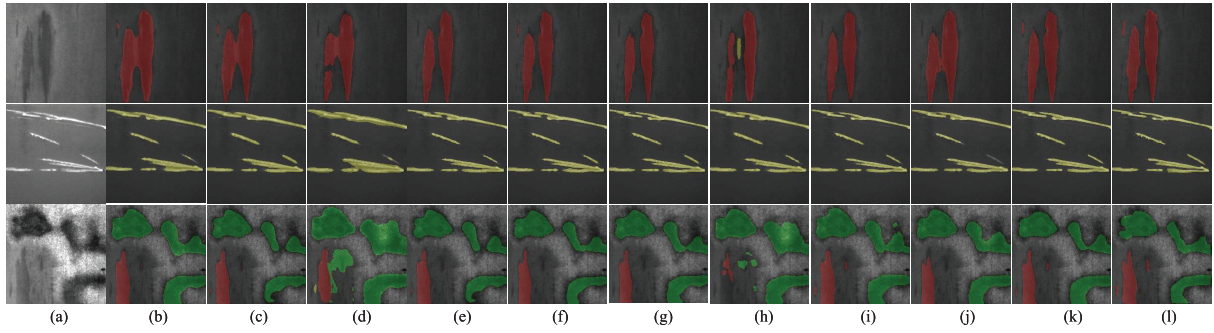


Fig. 5. Segmentation example of the NEU-Seg dataset. Each row denotes, in sequence, IN, SC, and PA, where the red, yellow, and green colors in the results denote the IN, SC, and PA, respectively. (a) Original image. (b) GSLU-Net. (c) PSPNet. (d) DNLNet. (e) CTNet. (f) PIDNet. (g) DMANet. (h) ISANet. (i) CRNet. (j) DDSNet. (k) Proposed. (l) Ground truth.

TABLE II
RESULTS OF DIFFERENT COMPARISON METHODS ON NEU-SEG DATASET

Methods	Precision					Recall					F1-score					IoU				
	BG	IN	PA	SC	Mean.	BG	IN	PA	SC	Mean.	BG	IN	PA	SC	Mean.	BG	IN	PA	SC	Mean.
GSLU-Net [18]	98.1%	82.0%	86.2%	89.2%	88.9%	98.1%	68.7%	92.4%	85.2%	86.1%	98.1%	74.8%	89.2%	87.1%	87.5%	96.3%	59.7%	80.5%	77.2%	78.4%
PSPNet [36]	97.9%	78.4%	93.0%	88.6%	89.5%	98.6%	73.9%	87.0%	86.2%	86.4%	98.2%	76.1%	89.9%	87.4%	87.9%	96.5%	61.4%	81.6%	77.6%	79.3%
DNLNet [37]	98.2%	82.8%	93.7%	88.4%	90.8%	98.7%	78.0%	89.3%	88.9%	88.7%	98.5%	80.3%	91.5%	88.7%	89.8%	97.0%	67.1%	84.3%	79.6%	82.0%
CTNet [38]	98.0%	79.8%	95.3%	90.6%	90.9%	98.8%	80.3%	87.5%	84.6%	87.8%	98.4%	80.1%	91.2%	87.5%	89.3%	96.7%	66.7%	83.9%	77.8%	81.3%
PIDNet [39]	98.0%	82.2%	92.7%	81.2%	88.5%	98.4%	74.6%	90.0%	88.9%	88.0%	98.2%	78.2%	91.3%	84.9%	88.2%	96.5%	64.7%	80.0%	73.6%	78.7%
DMANet [19]	97.9%	85.4%	93.4%	89.4%	91.5%	98.8%	67.6%	90.3%	84.6%	85.3%	98.3%	75.5%	91.8%	86.9%	88.1%	97.0%	62.9%	85.0%	78.2%	80.8%
ISANet [40]	97.9%	85.8%	94.0%	88.7%	91.6%	98.9%	69.6%	88.4%	87.2%	86.0%	98.4%	76.9%	91.1%	87.9%	88.6%	96.9%	64.0%	83.7%	78.4%	80.8%
CRNet [21]	98.0%	75.9%	93.0%	87.1%	88.5%	98.4%	75.0%	87.7%	87.5%	87.2%	98.2%	75.5%	90.3%	87.3%	87.8%	96.5%	60.6%	82.3%	77.5%	79.2%
DDSNet [20]	98.6%	88.6%	88.4%	79.7%	88.8%	98.1%	86.4%	86.3%	86.6%	89.4%	98.6%	87.5%	87.3%	83.0%	89.1%	97.3%	69.4%	76.4%	72.2%	78.8%
Proposed	98.7%	86.2%	94.3%	91.5%	92.7%	98.9%	83.8%	92.5%	91.3%	91.6%	98.8%	85.0%	93.4%	91.4%	92.2%	97.7%	73.9%	87.5%	84.2%	85.8%

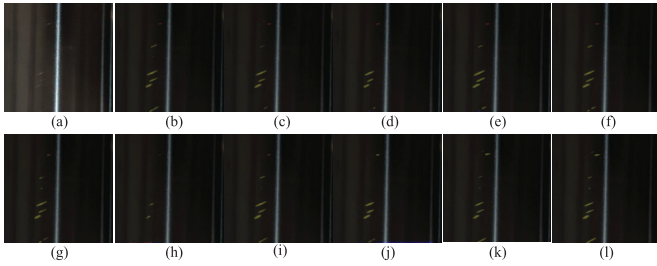


Fig. 6. Segmentation examples for SC defects in the CVC-Seg dataset, where yellow denotes the SC defects. (a) Original image. (b) GSLU-Net. (c) PSPNet. (d) DNLNet. (e) CTNet. (f) PIDNet. (g) DMANet. (h) ISANet. (i) CRNet. (j) DDSNet. (k) Proposed. (l) Ground truth.

methods in all evaluation metrics. Specifically, the proposed method achieves an average Precision of 92.7%, which is 2.8% higher than the comparison method, indicating that VICIM performs better in terms of classification accuracy. Furthermore, although the Recall for the IN class is 2.6% lower than DDSNet, the overall performance remains optimal, suggesting that the proposed method has an advantage in reducing false negatives. The average IoU of the proposed method is 5.9% higher than comparison methods, further demonstrating VICIM's superior localization accuracy. In summary, VICIM can effectively achieve high-precision defect detection.

2) *CVC-Seg Analysis*: Fig. 6 illustrates a comparison of the visual results of some of the CVC-Seg defect images detected. The main challenges of this dataset are the small

size of the defects and the low contrast between the background and the defects. It can be seen that the detection results of the proposed method are closer to the ground truth, while the comparison method shows more misdiagnosis and missed detections. Table III gives the quantitative comparison results of the accuracy of the proposed method with the other methods. In terms of Precision, the overall performance of the proposed method is better than the comparison methods by an average of 5.4%, indicating that it has a lower pixel misdiagnosis rate. Meanwhile, the *F1*-score of the proposed method is 4.7% higher on average, showing its advantage in balancing the misdiagnosis rate with the missed rate. More importantly, the mIoU of the proposed method reaches 73.2%, which is 5.6% higher than that of the comparison methods on average, indicating its significant advantage in localization accuracy.

3) *MFL-Seg Analysis*: Fig. 7 shows a visualization of the ground truth of the pipeline defects compared to the corresponding predictions. The main challenge of this dataset is the mutual coupling between defects and welds, which significantly increases the difficulty of the model to classify the coupled regions. As can be seen in Fig. 7, most comparison methods tend to misclassify welds as defects. In contrast, the proposed VICIM performs optimally in all performance indicators. Compared with other methods, VICIM can detect defective areas more completely. As indi-

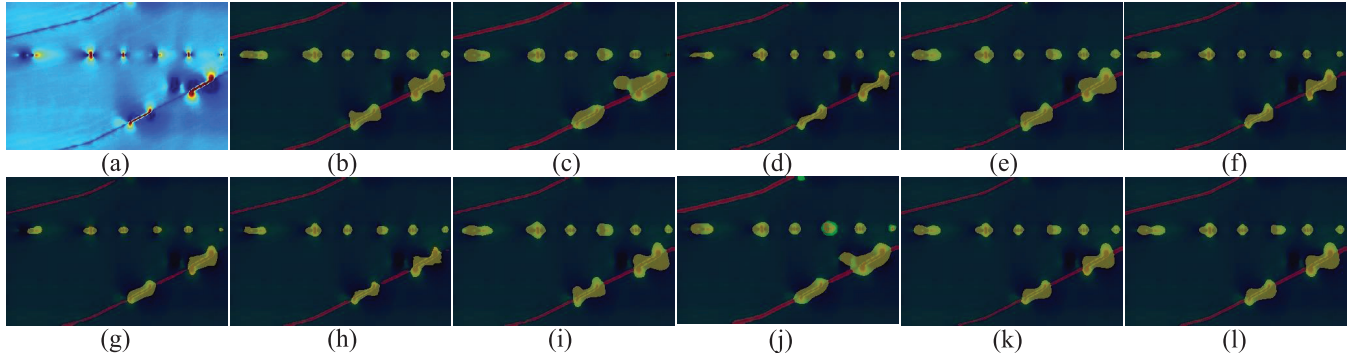


Fig. 7. Segmentation example of the MFL-Seg dataset. The red and yellow colors in the results denote SW and WD, respectively. (a) Original image. (b) GSLU-Net. (c) PSPNet. (d) DNLNet. (e) CTNet. (f) PIDNet. (g) DMANet. (h) ISANet. (i) CRNet. (j) DDSNet. (k) Proposed. (l) Ground truth.

TABLE III
RESULTS OF DIFFERENT COMPARISON METHODS ON CVC-SEG DATASET

Methods	Precision						Recall						F1-score						IoU					
	BG	IND	PM	SC	VI	Mean.	BG	IND	PM	SC	VI	Mean.	BG	IND	PM	SC	VI	Mean.	BG	IND	PM	SC	VI	Mean.
GSLU-Net [18]	99.5%	63.2%	82.1%	76.5%	86.9%	81.6%	99.5%	70.4%	80.4%	51.8%	87.9%	77.8%	99.5%	66.6%	81.2%	61.8%	87.4%	79.3%	99.0%	53.4%	68.4%	44.7%	75.4%	68.2%
PSPNet [36]	99.4%	73.5%	82.7%	76.5%	85.7%	83.6%	99.6%	72.3%	78.8%	55.0%	85.9%	78.3%	99.5%	72.9%	80.8%	64.0%	85.8%	80.6%	99.0%	57.4%	67.7%	47.0%	75.9%	69.4%
DNLNet [37]	99.3%	72.7%	80.4%	77.8%	87.7%	83.6%	99.6%	62.3%	73.5%	41.5%	82.6%	71.9%	99.4%	67.2%	76.8%	54.1%	85.1%	76.5%	98.9%	50.6%	62.4%	37.1%	74.0%	64.6%
CTNet [38]	99.3%	75.9%	81.6%	77.4%	87.2%	84.3%	99.6%	64.1%	78.2%	48.5%	84.0%	74.9%	99.5%	69.5%	79.9%	59.6%	85.6%	78.8%	99.0%	53.2%	66.5%	42.5%	77.0%	67.6%
PIDNet [39]	99.5%	72.6%	79.2%	74.5%	86.0%	82.4%	99.4%	70.1%	80.3%	55.9%	85.6%	78.3%	99.5%	71.3%	79.7%	63.8%	85.8%	80.0%	99.0%	55.4%	66.3%	46.9%	78.1%	69.1%
DMANet [19]	99.5%	77.1%	83.0%	81.2%	88.3%	85.8%	99.6%	70.9%	78.6%	52.8%	89.4%	78.3%	99.5%	75.4%	80.7%	64.0%	88.8%	81.7%	99.1%	60.5%	69.0%	47.0%	76.9%	70.5%
ISANet [40]	99.1%	64.8%	82.5%	77.8%	86.3%	82.1%	99.7%	50.8%	75.9%	40.1%	76.8%	68.7%	99.4%	57.0%	79.1%	52.9%	81.3%	73.9%	98.8%	39.8%	67.6%	36.0%	71.6%	62.7%
CRNet [21]	99.5%	66.1%	84.2%	76.3%	87.1%	82.6%	99.5%	70.7%	75.4%	51.9%	88.8%	77.3%	99.5%	68.3%	79.6%	61.8%	87.9%	80.0%	99.0%	55.6%	66.1%	44.7%	78.5%	68.8%
DDSNet [20]	99.2%	72.4%	82.6%	70.8%	83.9%	81.8%	99.4%	74.2%	73.8%	47.6%	83.1%	75.6%	99.4%	73.3%	78.0%	56.9%	83.5%	78.2%	98.7%	57.8%	63.9%	42.8%	74.7%	67.6%
Proposed	99.5%	84.3%	86.2%	84.5%	88.0%	88.5%	99.6%	73.3%	80.9%	56.2%	89.4%	79.9%	99.5%	78.4%	83.5%	67.5%	88.7%	83.5%	99.1%	64.5%	71.6%	50.9%	79.7%	73.2%

TABLE IV
RESULTS OF DIFFERENT COMPARISON METHODS ON MFL-SEG DATASET

Methods	Precision					Recall					F1-score					IoU				
	BG	SW	GW	DE	Mean.	BG	SW	GW	DE	Mean.	BG	SW	GW	DE	Mean.	BG	SW	GW	DE	Mean.
GSLU-Net [18]	99.2%	88.7%	92.9%	86.5%	91.8%	99.3%	91.2%	89.2%	77.7%	89.4%	99.3%	89.9%	91.0%	81.9%	90.5%	98.5%	82.6%	83.5%	69.3%	83.5%
PSPNet [36]	99.1%	87.5%	92.3%	84.9%	91.0%	99.2%	88.9%	88.1%	65.2%	85.4%	99.1%	88.2%	90.2%	73.8%	88.1%	98.3%	81.2%	82.1%	59.4%	80.3%
DNLNet [37]	99.0%	89.9%	93.1%	85.1%	91.8%	99.5%	89.6%	82.7%	77.1%	87.2%	99.2%	89.7%	91.5%	87.6%	92.0%	98.5%	83.1%	80.1%	67.9%	82.4%
CTNet [38]	99.3%	88.6%	91.2%	83.0%	90.5%	99.3%	87.9%	93.8%	85.4%	91.6%	99.3%	88.2%	92.5%	84.2%	91.1%	98.7%	81.1%	86.0%	72.6%	84.6%
PIDNet [39]	98.9%	88.2%	87.1%	85.8%	90.0%	99.3%	90.6%	76.7%	75.2%	85.5%	99.1%	89.4%	85.7%	80.2%	87.7%	98.4%	82.5%	75.0%	68.1%	81.0%
DMANet [19]	99.1%	88.9%	89.6%	84.0%	90.4%	99.2%	86.2%	93.4%	78.9%	89.4%	99.1%	87.5%	91.4%	81.4%	89.9%	98.5%	80.9%	84.2%	68.6%	83.1%
ISANet [40]	99.0%	87.9%	89.3%	85.8%	90.5%	99.3%	88.1%	91.3%	67.8%	86.6%	99.3%	88.0%	89.6%	75.7%	88.2%	98.5%	81.8%	84.4%	69.1%	83.5%
CRNet [21]	99.3%	91.3%	90.2%	83.4%	91.1%	99.3%	90.2%	93.7%	83.1%	91.6%	99.3%	90.7%	91.9%	83.2%	91.3%	98.6%	83.0%	85.1%	71.2%	84.5%
DDSNet [20]	99.0%	89.9%	92.0%	80.3%	90.3%	99.2%	92.0%	74.7%	85.9%	88.0%	99.1%	91.0%	82.5%	83.0%	88.9%	98.3%	83.4%	70.1%	71.0%	80.7%
Proposed	99.5%	89.5%	94.2%	86.7%	92.4%	99.3%	91.6%	95.2%	86.9%	93.3%	99.4%	90.5%	94.2%	86.8%	92.7%	98.7%	83.6%	89.1%	78.8%	86.7%

cated in Table IV, VICIM improves the average Recall and average IoU to 93.3% and 86.7%, which are 5.0% and 4.1% higher than the comparison methods, respectively. This indicates that the proposed method can realize more accurate pixel classification.

The reasons for the excellent performance of the proposed method can be summarized as follows.

- 1) At the feature level, the domain information is combined with the a priori knowledge of the generalized large model, enabling the vision encoder to focus on

defective regions that match the semantic descriptions (e.g., long, thin, irregular edges, etc.). This design effectively exploits potential defect characteristics hidden in the background, thus improving detection accuracy in complex scenes.

- 2) At the decision level, based on the decision-making principle of opposites and complementarities, the DSDM enhances the model's ability to discriminate edge regions, thus further ensuring the recognition accuracy of complex regions.

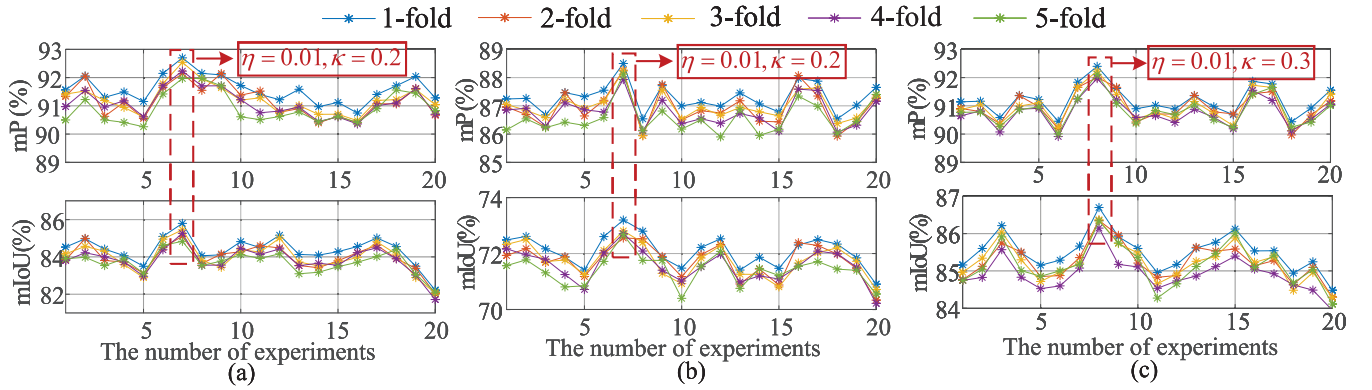


Fig. 8. Fivefold cross-validation results for hyperparameter selection. (a) Results of NEU-Seg. (b) Results of CVC-Seg. (c) Results of MFL-Seg.

D. Parameter Sensitivity Analysis

In this section, the optimal choice of hyperparameters (η and κ) is determined by exhaustive combinations and fivefold cross-validation experiments, where η is applied to balance the optimized language features, and κ is applied to balance the loss function of different terms. Similar to the work in [42], the mP and mIoU are utilized to evaluate the model performance. Multiple experimental setups are carried out using Microsoft's auto-tuning tool NNI to identify the optimal hyperparameter values. In terms of η , inspired by [31], η is a balance item less than 0.1 to maximize the retention of the original valid language information. In terms of κ , κ serves as a balancing term for the auxiliary loss function, and the value of κ usually does not exceed the loss term of the main task. Therefore, based on the above analysis and computational resource considerations, the range of hyperparameters is carefully defined as: $\eta \in \{0.001, 0.01, 0.05, 0.1\}$ and $\kappa \in \{0.1, 0.2, 0.3, 0.4, 0.5\}$. Naturally, the experiments are carried out sequentially on each of the three industrial datasets to determine the optimal parameter combinations. The experimental results are illustrated in Fig. 8, where mP and mIoU reach their optimal values in the seventh, seventh, and eighth experiments on the three datasets, respectively. Therefore, the hyperparameters for the three datasets are set to $\{0.01, 0.2\}$, $\{0.01, 0.2\}$, and $\{0.01, 0.3\}$. Overall, as the moderator of the loss function, η is more sensitive to the distribution of the dataset. When the value of κ is too large, the auxiliary task dominates the gradient of the main task too much, causing the model optimization to deviate from the main segmentation task, which may instead lead to a decrease in the accuracy of the model. Therefore, κ is usually initialized to a smaller value to ensure the stability and accuracy of the model. In addition, from the results of multiple rounds of cross-validation, the overall performance of the model fluctuates less under different experimental configurations, which further verifies that the proposed VLCIM has good robustness under hyperparameter variations.

E. Model Complexity Analysis

In this section, the computational complexity of the model is analyzed. The asymptotic complexity of the proposed method is primarily determined by the Transformer-based feature

TABLE V

PARAMETERS AND TEST TIME OF DIFFERENT METHODS

Methods	Parameters(M)	Test time(s/batch)
GSLU-Net [18]	52.13	0.023
PSPNet [36]	46.60	0.016
DNLNet [37]	47.65	0.014
CTNet [38]	46.53	0.017
PIDNet [39]	37.31	0.010
DMANet [19]	42.22	0.015
ISANet [40]	35.34	0.013
CRNet [21]	60.45	0.029
DDNet [20]	56.34	0.026
Proposed	98.21	0.038

TABLE VI

RESULTS OF ABLATION EXPERIMENTS ON MFE

Model	Precision			IoU		
	IN	PA	SC	IN	PA	SC
w/o EC	85.4%	92.6%	90.5%	73.2%	86.2%	83.4%
w/o AL	85.6%	92.1%	89.3%	71.4%	85.8%	83.2%
ResNet50	84.2%	92.7%	88.8%	69.1%	84.3%	79.3%
ResNeSt	85.9%	93.4%	89.1%	72.9%	86.9%	83.5%
Proposed	86.2%	94.3%	91.5%	73.9%	87.5%	84.2%

extraction module and can be approximated as $O(n^2)$, where n denotes the total number of input pixels. The number of parameters and the per-batch inference time for different methods are summarized in Table V. Although the proposed method is not the most efficient in terms of parameter count and inference speed, it achieves a per-batch inference time of 0.038 s, which is sufficient to meet the requirements of current second-level online analysis tasks and offline data processing scenarios. Moreover, in comparison with methods that have fewer parameters, the proposed approach exhibits no significant difference in inference time, yet delivers superior detection accuracy. This demonstrates its higher practical applicability and overall performance advantage.

F. Ablation Studies

Ablation studies are conducted to determine the importance of each component on the NEU-Seg dataset, which mainly focus on MFE, progressive cyclic interactive learning (PCIL), and DSDM. The details are as follows.

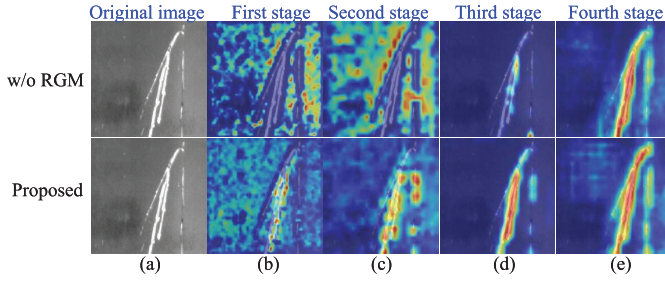


Fig. 9. Results of the heatmap visualization of the attention at each stage of the visual feature extraction network, where the red area indicates the model's focus on the defect. (a) Original image. (b) Heat map of the first stage. (c) Heat map of the second stage. (d) Heat map of the third stage. (e) Heat map of the fourth stage.

TABLE VII
RESULTS OF ABLATION EXPERIMENTS ON PCIL

Model	Precision			IoU		
	IN	PA	SC	IN	PA	SC
w/o CMI	85.4%	92.4%	90.4%	73.1%	84.0%	82.8%
w/o RGM	84.7%	93.1%	85.0%	71.8%	83.3%	80.9%
Proposed	86.2%	94.3%	91.5%	73.9%	87.5%	84.2%

1) *Analysis of MFE*: In the MFE phase, this study focuses on verifying the effectiveness of EC, the AL, and different vision encoders. As shown in Table VI, when the EC mechanism is not introduced, the detection performance of the model decreases, with mean average precision (mAP) and average IoU decreasing by 1.2% and 0.9%, respectively. Meanwhile, when AL is not applied, the overall performance of the model is degraded. The main reason is that the defect text description generated by the large model deviates from the defect attributes under actual working conditions, which weakens the guiding effect of the text on the visual features and affects the model's ability to focus on the key information of the defects, which, in turn, leads to the decrease in accuracy. In addition, when the Transformer-based vision encoder is replaced with ResNet and ResNeSt, the Precision and IoU of the model are reduced. This indicates that AL is crucial for fine-tuning the language prior. In addition, compared to CNNs, the Transformer-based vision coder can better maintain semantic consistency with the language encoder, thus effectively improving the overall performance of the model.

2) *Analysis of PCIL*: PCIL is designed to achieve the joint optimization of vision information and language information, which consists of CMI and RGM. From Table VII, it can be observed that the performance of the model is degraded without any of the modules, and in particular, the average Precision and the average IoU of the model are decreased by 3.1% and 3.2%, respectively, when RGM is not applied. Fig. 9 shows the heatmap visualization of attention at each stage works of the vision encoder. It can be seen that RGM enables the model to pay better attention to defects. This suggests that the progressive guidance of language information enables the model to better focus on the defect information, which, in turn, safeguards the detection accuracy of the model.

3) *Analysis of DSDM*: DSDM is designed to mine defect information from complementary perspectives and achieve

TABLE VIII
RESULTS OF ABLATION EXPERIMENTS ON DSDM

Model	Precision			IoU		
	IN	PA	SC	IN	PA	SC
w/o FV	84.9%	92.8%	87.9%	71.5%	86.3%	83.0%
w/o BV	85.3%	93.6%	89.1%	72.8%	86.9%	83.1%
Proposed	86.2%	94.3%	91.5%	73.9%	87.5%	84.2%

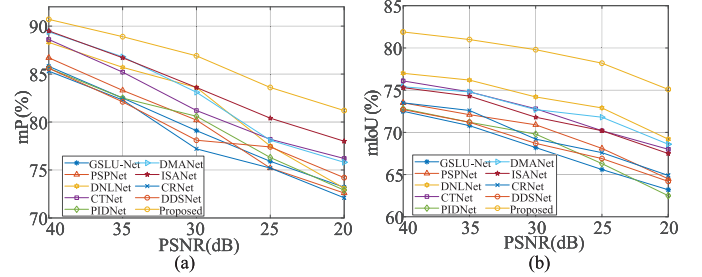


Fig. 10. Experimental results at different PSNRs. (a) mP for the NEU-Seg dataset. (b) mIoU for the NEU-Seg dataset.

defect detection. In order to verify the necessity of different DSDM branches (FV and BV), a series of ablation experiments is conducted. The experimental results are shown in Table VIII, and the lack of any view will lead to performance degradation. Specifically, the detection and localization accuracy decrease by an average of 2.1% and 1.6%, respectively, while FV is missing. These experimental results demonstrate the crucial role of DSDM in defect detection. Overall, the proposed detection mechanism achieves multilevel decision-making, leading to more precise detection outcomes.

G. Noise Robustness Verification

To verify the robustness of the proposed method, this section designs several sets of noise interference experiments based on the NEU-Seg dataset, where Gaussian white noise with different intensities is added to the original images. To comprehensively evaluate the model performance, the mP and mIoU are selected as the evaluation metrics, where only the detection results of various types of defects are calculated, ignoring the influence of the background region. The experimental results are shown in Fig. 10. It can be found that the performance of all methods decreases with the enhancement of noise, but the proposed method still outperforms the other compared methods. Specifically, the mP is still above 85% when the experimental results at different peak signal-to-noise ratio (PSNR) are higher than 30 dB. In addition, in terms of boundary localization, as the PSNR decreases, the IoU of the proposed method only decreases by 6.8%, which is significantly better than the other methods, indicating that the proposed method can have better segmentation boundaries even under noise interference.

The reasons for the better robustness can be summarized as follows.

- 1) The proposed progressive cyclic interaction learning realizes the deep modulation of vision features by language information and ensures the effectiveness of deep vision feature extraction, thus weakening the impact

of noise on the model's performance. In addition, by introducing language information, the model is guided to establish global semantic constraints during feature extraction, so that it can still make accurate judgments based on semantic consistency in the face of local noise interference, thus significantly improving robustness.

- 2) Compared with existing methods that typically rely on a single network for decision-making, the proposed DSDM enables complementary decision fusion, effectively mitigating the influence of individual noise inputs and single-point decisions. As a result, the proposed method demonstrates superior robustness.

V. CONCLUSION

This article has proposed a novel industrial defect detection method named the VLCIM, including MFE, progressive cyclic interaction learning, and DSD. The core innovation lies in the construction of a two-way interaction mechanism between domain-specific language features and vision representations. First, expert experience is combined with a generic large model to obtain domain-specific language information, which is applied to guide the extraction of vision features. Subsequently, the language information optimizes the semantic features generated by the vision coder layer by layer so that it can focus on the defective features related to the language description. Meanwhile, the deeper vision semantic information can optimize the language embedding through CMI in reverse. Finally, a DSDM is designed to fully exploit the discriminative information of defects and improve the accuracy of complex defect detection. In contrast to existing methods that rely on static linguistic cues or unidirectional guidance, the proposed method emphasizes cyclic and gradual mutual augmentation between patterns to produce context-aware feature representations. The experimental results on three industrial datasets demonstrate that VLCIM consistently outperforms SOTA methods on three industrial datasets, achieving IoU improvements of 5.9%, 5.6%, and 4.1%, respectively. Additionally, it achieves performance gains of up to 5.4% in mP.

Although the proposed method achieves high-precision defect detection, it requires the involvement of experts and has a high computational complexity. Future work will focus on developing lightweight and end-to-end vision-language defect detection models to enhance computational efficiency.

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